

NAVEENRAJ

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PROFESSIONAL SUMMARY

Results-driven Software Engineer with extensive experience designing and deploying enterprise-scale AI systems, backend microservices, and big data analytical pipelines. Specialized in designing cloud-agnostic architectures, multi-tenant databases, real-time RAG processing frameworks, and automated infrastructure provisioning. Adept at transforming complex datasets into actionable natural language interfaces using advanced LLM orchestration, intelligent automation adapters, and ML-driven predictive forecasting.

CORE COMPETENCIES & SKILLS

- AI & Machine Learning:** LLM Integration (AWS Bedrock Claude/Nova, Azure OpenAI), Agentic Multi-Agent Frameworks, RAG Pipelines, Vector Search, MLflow Tracking, Time-Series Forecasting.
- Languages & Web Frameworks:** Python (FastAPI, Flask, PySpark), Kotlin, Dart, JavaScript, PHP, SQL, HTML/CSS.[cite: 2]
- Data Engineering & Cloud Platforms:** Databricks (Delta Lake, Vector Search), AWS (Lambda, API Gateway, SQS FIFO, Secrets Manager, S3, SES), Azure (AKS, Function Apps, Service Bus, Key Vault, Cosmos DB Gremlin API).
- Databases & Tools:** PostgreSQL, MongoDB, MySQL, Microsoft SQL Server, Room DB, Docker, Terraform, Git.[cite: 2]
- Architecture & Patterns:** Microservices Architecture, Multi-Tenant Database Design, Platform Adapter Pattern, Dual-Trigger Async Processing, AST Parsing.

PROFESSIONAL EXPERIENCE

Software Engineer – AI & Backend Systems	January 2024 – Present	[cite: 2]
<i>Information Technology & Software Services Domain [cite: 2]</i>		<i>Chennai, TN</i>
- Architect and implement cloud-native microservices and big data analytical applications leveraging Python, FastAPI, Flask, and Databricks. - Design robust multi-tenant architectures supporting real-time database schema switching and production migrations across heterogeneous environments. - Lead the transition of legacy enterprise operations into automated, AI-driven workflows utilizing agentic design patterns and generative modeling. - Standardize backend components by developing reusable structural adapter interfaces and implementing comprehensive asynchronous validation patterns.		
Junior Developer & Intern	July 2023 – January 2024	[cite: 2]
<i>Information Technology & Software Services Domain [cite: 2]</i>		<i>Chennai, TN</i>

- Built scalable components using structural design patterns and integrated third-party remote communication APIs.[cite: 2]
- Developed specialized translation and data synchronization layers to safely bridge legacy backends with modern entry-point engines.[cite: 2]
- Completed advanced technical training covering application lifecycle states, secure network intent handling, and core responsive programming concepts.[cite: 2]

TECHNICAL PROJECTS

AI/BI Querying Engine & Predictive Forecasting Model

Current Project (2 Months)

Technical Stack: Databricks Genie, Databricks AI/BI Dashboards, MLflow, PySpark, Delta Lake

- **Genie Configuration:** Provisioned and fine-tuned a Databricks Genie Space to seamlessly convert natural language business queries into precise, validated SQL executions against massive database schemas.
- **Predictive Machine Learning:** Developed and deployed a robust sales and inventory forecasting model using MLflow to track parameters, handle versioning, and register models.
- **Data Visualization:** Built dynamic Databricks AI/BI dashboards to provide stakeholders with immediate, automated visual insights from real-time and predicted retail metrics.

Platform Orchestrator – Multi-Cloud Chatbot Deployment Automation

Project Duration: 2 Months

Technical Stack: Python 3.11, FastAPI, Terraform, AWS Lambda, SQS FIFO, Azure Functions, GCP Functions, Databricks Genie

- **Dual-Trigger Pattern:** Implemented an asynchronous webhook receiver loop using AWS Lambda and SQS FIFO queues to completely decouple ingestion from long-running analytical executions.
- **Platform Adapter Pattern:** Designed an abstract interface layer (SlackAdapter / TeamsAdapter) to handle discrete payload validations and internal structural mapping across diverse communication channels.
- **Cloud-Agnostic Design:** Created reusable multi-cloud Terraform modules to deploy identical business execution layers uniformly across AWS, Azure, and GCP.
- **Automated Visualization:** Programmed a heuristic data-shape detection module that automatically parses Databricks query row counts and column profiles into custom visual charts.

AI-Powered Service Management Platform

Project Duration: 10 Months

Technical Stack: Flask REST API, PySpark, AWS Bedrock, Google Maps API, AWS SES, PostgreSQL, MySQL

- **AI-Driven Scheduling & Allocation:** Built an intelligent job-scheduling engine that leverages AWS Bedrock to estimate project durations, computes transit travel times via Google Maps API, and evaluates real-time truck availability.
- **Dynamic Pricing Engine:** Engineered a math-driven dynamic pricing optimizer tracking real-time fleet utilization and YoY demand parameters to execute automated price modifications (+5% increase to -5% decrease constraints).
- **Multi-Tenant Complexity:** Resolved dynamic schema routing layers based on runtime system attributes, encapsulating database pool switching across simultaneously running database versions.
- **Automated Notifications (SPNS):** Built a multi-system scheduled payment tracker that dynamically binds database contexts to extract overdue profiles, automatically generating customer-specific HTML summaries and custom Excel attachments via AWS SES.

Location Resolution System – PySpark Pipeline

Project Duration: 2 Months

Technical Stack: Databricks (PySpark, Delta Lake), Databricks Vector Search, Python ThreadPoolExecutor, Serper API

- **Deterministic & Vector Search (Tier 1):** Configured multi-threaded PySpark workers (200 threads) to process exact name lookups against reference datasets, failing over to a 9-table Databricks Vector Search RAG pipeline with a strict threshold score ≥ 0.8 .
- **Web Enrichment & LLM (Tier 2):** Engineered a secondary fallback pipeline utilizing 20 worker threads that dynamically scrapes company websites or executes automated Google Searches via Serper API, passing extracted text to an LLM to finalize unresolved records.

Advanced Enterprise Code Intelligence Engine

Project Duration: 6 Months

Technical Stack: Python FastAPI, Azure OpenAI, AWS Bedrock, Azure Cosmos DB (Gremlin Graph API), MongoDB, Tree-Sitter

- **Multi-LLM Integration:** Wired multi-LLM chatbot features leveraging AWS Bedrock Claude and Azure OpenAI to deliver highly contextual code reasoning.
- **Abstract Syntax Tree Parsing:** Implemented localized language code parsing utilizing tree-sitter to break down raw source files into hierarchical tree frameworks.
- **Call Graph Generation:** Developed a structural engine to generate comprehensive function call dependency networks, tracing relationships from single lines up to root architectures.
- **Performance Refactoring:** Managed Microsoft SQL to PostgreSQL database schema migrations while building async logging subsystems to eliminate thread blocks during high workload analysis.

AI-Powered Organizational Productivity Engine

Project Duration: 6 Months

Technical Stack: Python Flask, Azure Kubernetes Service (AKS), Azure OpenAI, MongoDB Vector Store

- **RAG Engineering:** Programmed an end-to-end document processing pipeline that parses unstructured files, creates semantic chunks, and stores text embeddings within a MongoDB vector store for semantic retrieval.
- **Prompt Optimization Module:** Developed an intelligent context module that dynamically alters and enriches prompt payloads based on live user context and parameters.
- **Asynchronous Streaming:** Configured an optimized response streaming infrastructure to enable real-time, low-latency conversational output deliveries.
- **Containerized Orchestration:** Migrated the backend microservices from a serverless architecture into containerized deployments managed via Azure Kubernetes Service (AKS).

EDUCATION

Bachelor of Engineering (B.E.) in Electrical and Electronics Engineering [cite: 2] August 2019 – May 2023 [cite: 2]

CK College of Engineering and Technology, Cuddalore, TN | CGPA: 8.7/10 [cite: 2]